

Soumyadeep Bhowmik

📞 9932553596 [in soumyadeep-bhowmik](#) [@soumyacodes007](#)

EDUCATION

Narula Institute of Technology | West Bengal, India

Aug 2024 – May 2028

Bachelor of Technology in Computer Science and Engineering SGPA: 8.9

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Solidity, SQL, C++, Rust (Basics)

AI/ML: LangChain, RAG, Graph RAG, LLM Fine-tuning (Unsloth), RL, MCP, A2A, HuggingFace, ONNX, Autograd

Tools/Cloud: Docker, Git, AWS, Postman, Vercel, Neon Postgres, Drizzle ORM, Vector DBs (Chroma, Pinecone, FAISS)

WORK EXPERIENCE

AI Agent Engineer | *Alchemyst AI* | Remote, Bengaluru

May 2025 – Jul 2025

- Developed AI agents transforming natural-language prompts into automated 2D marketing videos and product demos.
- Integrated ElevenLabs for voice synthesis and FFmpeg for end-to-end automated media processing pipelines.
- Optimized agentic workflows for RAG-based context injection, ensuring video scripts remained brand-consistent.
- Built scalable Streamlit interfaces for prompt-to-video workflows and containerized solutions using Docker for portable deployment.

KEY PROJECTS

GhostContext — Decentralized AI Knowledge Marketplace

[Live Demo](#) | [GitHub](#)

- Architected a client-side RAG marketplace using Walrus (Decentralized storage) and WebLLM for local inference.
- Implemented AES-256-GCM encryption with NFT-gated decryption rights enforced via Sui Move smart contracts.
- Integrated WeInfer engine achieving 3.76x faster browser inference on Llama-3/Phi-3 via WebGPU acceleration.
- Built a token-gated query system allowing experts to monetize datasets without compromising data sovereignty.

ONNX Deployment Surgeon Gym — OpenEnv RL Environment

[HF Spaces](#) | [GitHub](#)

- Built an RL environment for agents to diagnose/repair broken ONNX bundles using real session validation.
- Designed a severity-weighted reward system for multi-step repair tasks across Vision and NLP models (MobileNet, ResNet).
- Implemented deterministic curriculum learning profiles to ensure reproducible agent training across 3 distinct task families.
- Deployed as a containerized API on Hugging Face Spaces with a web-based model inspection and repair interface.

DailyRoutine — AI-Powered Accountability Habit Tracker

[Live Demo](#) | [GitHub](#)

- Built a multi-agent system: Agent A (Auditor) scans streaks; Agent B (Enforcer) generates personalized guilt-trip reminders.
- Developed "Schrödinger's Warning," an AI deterrent that uses personal journal sentiment to discourage habit deletion.
- Integrated Vercel AI SDK for low-latency streaming of AI-generated mood analysis and habit recommendations.
- Tech Stack: Next.js 15, Drizzle, Neon, Groq (Llama-3), Vercel AI SDK, TailwindCSS, Clerk Auth.

SIDE PROJECTS

- **ARGO RAG:** Conversational AI for analyzing oceanographic data using Gemini LLM and ChromaDB ([GitHub](#)).
- **Mini Pytorch From Scratch:** Built a custom Autograd engine, Tensors, and CNN/RNN layers using NumPy ([GitHub](#)).
- **own_llm:** Pure implementation of the "Attention is All You Need" Transformer architecture from scratch ([GitHub](#)).
- **A2A Agent:** Implemented how agents communicate with each other using googles a2a[LinkedIn](#).

ACHIEVEMENTS & LEADERSHIP

- **Hackathon Wins:** Won **17 hackathons** across AI, blockchain, and hardware domains nationally and internationally.
- **IISc Bangalore:** Won Ideathon at the Indian Institute of Science, Bangalore for high-impact research innovation.
- **IIT Kharagpur:** Secured 1st Place in the blockchain track at IIT KGP national hackathon against top-tier teams.
- **JU Srijan:** Won the hardware hackathon at Jadavpur University's Srijan tech fest for robotics integration.
- **Blockchain Protocols:** Won track prizes for Celo, Flare, and Stellar blockchain hackathons for Web3 dApp development.
- **MLH Hackathons:** Twice winner of **Major League Hacking (MLH)** international hackathons in competitive rounds.
- **ML Competitions:** Secured top ranks in **3 machine learning competitions** on public leaderboards and Kaggle.
- **Open Source:** Developed and published a public AI/ML SDK with **500+ downloads** on the package registry.
- **AI/ML Speaker:** Conducted a hands-on technical workshop for **200+ students** on building production-grade RAG.
- **Community Building:** Co-led a tech community from **0 to 3,000 members** in 4 months via structured mentorship.
- **Entrepreneurship:** Founded a startup, secured initial funding, and executed the full product development and GTM cycle.